

ENGINEERING SPECIFICATIONS & INSTALLATION

Molitron Odor Abatement System

Model MOAS — Atomized Odor & Smoke Neutralization



ISOMETRIC INSTALLATION VIEW — CABINET ON WALL, TUBING RUN TO NOZZLE PLATE ON EXHAUST DUCT

Capacity	Up to 6,000 CFM per system (two nozzles)
Electrical	115 VAC, 15 A dedicated circuit
Water	¼" copper cold water supply, 80 PSI
Cabinet	18 Ga stainless, 24"W × 32"H × 8"D, 100 lbs
Listing	ETL Listed (UL 197; UL 710 Cl. 43)

System Overview

The MOAS is a totally self-contained neutralization system: every component except the two remote misting nozzles lives in one wall-mounted stainless cabinet.

It runs standalone or alongside the Enviro-Clean Air Scrubber and Enviro-Pak Filter Assembly, and only operates when the exhaust fan runs (interlock required — page 4).

Inside the Cabinet



REF	COMPONENT
A	IDEC time controller enclosure
B	24 V Odor Gone solution pump
C	115 V GAST air compressor
D	24 V solenoid valve & solution injector
E	Mixing tank
F	One-gallon Odor Gone concentrate
G	1" conduit connection — tubing to misting nozzles
H	Cooling fan
J	Fresh air inlet
K	Cam latch handle
P	115 VAC, 15 A supply connection
T	Air & solution lines to nozzles
W	¼" cold water supply connection

MOUNTING

Wall-mounted via 16" keyhole hangers. A 1¼" hole for solution and alarm tubes is located 4" from the back side, 3" from the right side (field-drilled on existing units).

Misting Nozzle Installation

Two misting nozzles are furnished factory-mounted to a 4" square stainless steel plate with 8 mounting holes, installed on the **outside** of the kitchen exhaust duct.

Procedure

1. **Position for dwell time.** Locate nozzles as far as possible from the point of atmospheric discharge — more duct run after the nozzle means more complete neutralization.
2. **Cut** a 1½" hole in the side wall of the exhaust duct.
3. **Weld** steel studs to the duct side wall matching the holes in the stainless mounting plate.
4. **Mount** the nozzle assembly with the spray nozzle centered over the 1½" hole.
5. **Route lines** from the cabinet: ¼" **RED** solution line and ⅜" **BLACK** compressed air line (two sets of tubing and all fittings furnished). **Maximum line length: 100 feet.**



DUCT SECTION WITH NOZZLE PLATE DETAIL + LINE ROUTING TO
CABINET, COLOR-CODED RED / BLACK

SERVICEABILITY

The nozzle-and-plate assembly removes from the duct side wall as a unit — no duct entry required for service.

Electrical Installation

CONNECTION	REQUIREMENT	BY
Power	115 VAC, 15 A dedicated circuit to cabinet connection P	Contractor
Water	¼" copper cold water supply, 80 PSI, to connection W	Contractor
Interlock	MOAS / exhaust-fan interlock control relay (below)	Contractor

Required: Exhaust Fan Interlock

The MOAS must only run when the exhaust fan runs. The interlock relay energizes MOAS power only when the fan circuit is live, preventing solution use with no airflow.

- **Preferred source:** the 120 V hood-switch circuit that activates the exhaust fan's magnetic motor starter coil.
- **208-230 V fans:** use one of the 120 V phases.



INTERLOCK RELAY WIRING DIAGRAM — WHT/BLK 120 VAC ·
WHT/BLU 24 V · WHT/YEL COMMON · LINE/LOAD N.O. CONTACTS

Solution System & Calibration

Odor Gone concentrate is automatically diluted by the injector and mixed in the cabinet's mixing tank before atomization.

Tuning performance: system output is increased or decreased by adjusting the dilution ratio set point on the brass solution injector at the cold-water supply. Start-up and calibration must follow the installation manual — set once to the site's odor load, then leave it alone.

Optional 10-Gallon Solution Container

For high-usage sites, add the wall-mounted reserve container:

Capacity	10 gallons (fill line marked)
Construction	16 Ga stainless, 24"W × 12"H × 8"D
Weight	20 lbs empty / 100 lbs full
Filling	Sliding lid with angle grip
Tubing	½" × 30" tube containing solution + alarm tubes (custom lengths at order; keep runs ≤ 8 ft)

NEVER RUN DRY WITHOUT KNOWING

An audible alarm sounds when less than 2 gallons remain — refills happen on your schedule, not after the odor complaints resume.



SOLUTION CONTAINER — FRONT VIEW WITH FILL LINE AND 2-GALLON ALARM THRESHOLD MARKED

Compliance



ETL Listed — tested to **UL 197** (Commercial Electric Cooking Appliances) and **UL 710 Clause 43** (Fire & Leakage Test of Fittings for Hood Penetrations).

Installer's Quick Checklist

- ☐ Cabinet wall-mounted; keyhole hangers secure
- ☐ Nozzle plate welded/mounted, spray nozzle centered on 1½" hole
- ☐ Nozzles located for maximum duct run before discharge
- ☐ Solution (red) and air (black) lines routed, ≤ 100 ft
- ☐ 115 VAC 15 A circuit landed; interlock relay wired to fan circuit
- ☐ ¼" cold water at 80 PSI connected
- ☐ Dilution ratio calibrated per installation manual
- ☐ Optional container mounted, alarm tube connected, filled

Ready to install. Ready to help.

- **Installation Support** 303.969.8888
- **Order Odor Gone Concentrate** cleanair@molitron.com
- **Manuals & Drawings** molitron.com

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